Lancaster University

2017 EXAMINATIONS

PART II – Third Year

MATHEMATICS & STATISTICS

Math 336: Multivariate Statistics in Machine Learning **$1\frac{1}{2}$ hours**

You should answer ALL SECTION A questions and ONE SECTION B question. If you answer more than one question from Section B then the best question will count. There is a total of 70 marks. You may only use the calculator that has been provided.

SECTION A

A1. You have been hired as a data scientist to help Boyds Bank automate its decision process for credit card applications. You have access to their historical data composed of $N > 10000$ credit card applications along with a binary (yes/no) decision for each application. Each credit card application contains the following information per applicant, age, gender, annual salary, current debt.

(a) Express this task in the framework of learning from data by specifying the input space $\mathcal{X}$, output space $\mathcal{Y}$, and the specifics of the dataset to be learned from. [6]

(b) Is this a supervised or an unsupervised learning problem? Justify your answer. [3]

(c) Since you are an expert in the field, Boyds Bank is assuming that after running your analysis you will provide a hypothesis that will definitely approximate the target function f well on new, unseen, credit card applications. Is this expectation correct? If not, what is the correct expectation? Justify your answer. [6]

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Question 1 continued

(d) Suppose that instead of a binary label, each data-point has a corresponding real-valued credit limit. If the bank's objective is to automate the task of credit limit assignment,

i. Which one of the following algorithms should be used? Justify your answer. [3]

* Perceptron Learning Algorithm.
* Linear Regression.

ii. Is this modified problem a supervised learning or an unsupervised learning problem? Justify your answer. [2]

A2. (a) What is overfitting? Give an example to show this phenomenon. [5]

(b) We are given a dataset corresponding to an unknown, potentially highly non-linear, real-valued target function $f : \mathbb{R} \to \mathbb{R}$. We wish to fit the data using one of the following models.

$\mathcal{H}_0$: is the set of all lines of the form $h(x) = b,\ b \in \mathbb{R}$

$\mathcal{H}_1$: is the set of all lines of the form $h(x) = ax + b,\ a, b \in \mathbb{R}$

It is known that $\mathcal{H}_0$ has bias equal to 0.5 and variance equal to 0.25, while $\mathcal{H}_1$ has bias equal to 0.21 and variance equal to 1.69. Given this information, which one of the two model classes is best suited for fitting the dataset at hand? [5]

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SECTION A continued

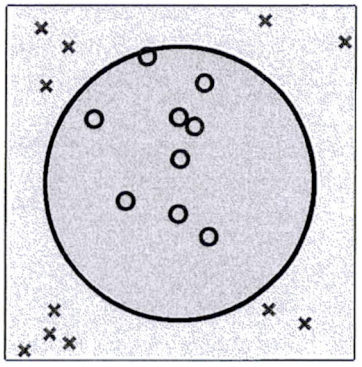
A3. Consider a binary classification problem with training set $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)$, where $\mathbf{x}_i \in \mathbb{R}^2$ and $y_i \in \{+1, -1\}$ for $i \in 1..N$.

(a) Consider the Perceptron Learning Algorithm (PLA) applied to this dataset. Let $\mathbf{w}(t)$ denote the PLA weight vector at iteration t. It is known that the point $\mathbf{x}_1$ has been *misclassified* by $\mathbf{w}(t)$.

i. Assuming that the data are linearly separable, state whether $y_1 \mathbf{w}^T(t)\mathbf{x}_1$ is positive or negative. [2]

ii. If $(\mathbf{x}_1, y_1)$ is selected at iteration t, how would you expect the PLA weight vector to be updated? [**Hint:** The answer should be of the form $\mathbf{w}(t+1) \leftarrow \ldots.$] [3]

(b) Given the below sample plot of the training data, where $\times$ denotes positive training examples (with label $+1$) and $\circ$ denotes negative training examples (with label -1), would you expect the Perceptron Learning Algorithm (PLA) to terminate in a finite number of iterations? Justify your answer. [3]



(d) Suppose we use a finite model class $\mathcal{H}$ for this classification problem. State without justification what the maximum value of the VC-dimension of $\mathcal{H}$ would be. [2]

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SECTION B

B1. Boogle Inc. has decided to improve the quality of the spam filter incorporated in their celebrated e-mail service. They have a dataset $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)$ of N e-mails, where each e-mail is represented as a 4-dimensional real vector $\mathbf{x}_i \in \mathbb{R}^4$, $i = 1..N$ and is given a binary label $y_i \in \{+1, -1\}$.

(a) Which one of the following two algorithms is appropriate for solving this problem? Justify and give a step-by-step description of your choice. [6]

i. Perceptron Learning Algorithm.

ii. Linear Regression.

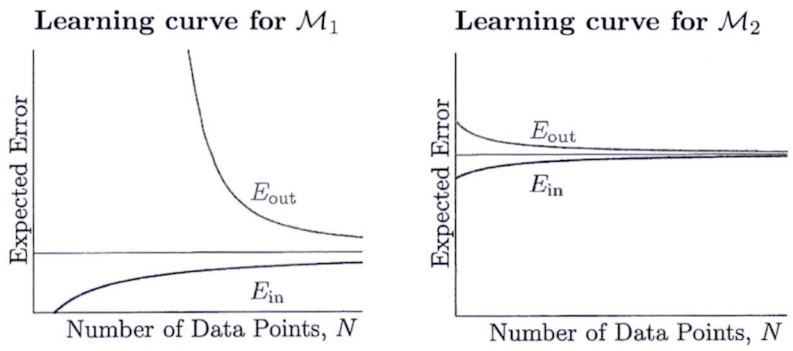
(b) Suppose that instead of a binary (spam/non-spam) decision, Boogle is interested in calculating the *probability* for a given e-mail to be spam. Propose a method to achieve this objective. [6]

(c) After much analysis, Boogle's data scientists are at a final decision stage where they need to select between two hypothesis classes, $\mathcal{H}_1$ and $\mathcal{H}_2$ from which they will choose a classifier to train.

i. Assume that the number N of data-points is large, e.g. $N > 100000$. As senior data scientist on the project, explain how you would advise your team to elect between the two model classes. [6]

ii. Assume that N is small, e.g. $N < 100$. Explain the trade-off between the size of the validation set, and the size of the training set. How would you modify your approach above to reliably choose the correct hypothesis class in this case? [6]

(d) Consider two model classes $\mathcal{M}_1$ and $\mathcal{M}_2$ with learning curves given below.



Boogle has asked the data science team to keep it simple! Therefore, it is required to choose the **simpler** of the two model classes for this classification problem.

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SECTION B continued

Question B1 continued

Which model class $\mathcal{M}_1$ or $\mathcal{M}_2$ should be selected under this condition? Justify your answer. [6]

B2. (a) Define the following concepts in the context of VC-analysis. [6]

i. Shattering

ii. Growth function

iii. VC-dimension

(b) We have a set of 4 points $\{x_1, \ldots, x_4\}$. It is known that a hypothesis set $\mathcal{H}$ implements a total of $M = 14$ dichotomies **on these** 4 **points**. Can we conclude that 4 is a break-point for $\mathcal{H}$? Justify your answer. [5]

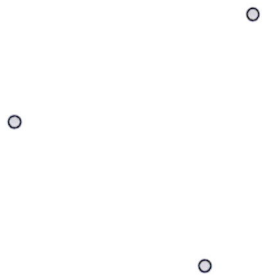
(c) Let $\mathcal{H}$ be a hypothesis set. It is known that for **every set of** 4 **points** $x_1, \ldots, x_4$, $\mathcal{H}$ implements $M < 16$ dichotomies. Can we conclude that 4 is a break-point for $\mathcal{H}$? Justify your answer. [5]

(d) State whether or not the following statement is true. [4]

If k is a break-point for $\mathcal{H}$ then all $k' \geq k$ are also break-points for $\mathcal{H}$.

(e) Give an example of a hypothesis set $\mathcal{H}$ with infinite VC-dimension, i.e. $d_{\text{VC}}(\mathcal{H}) = \infty$. [6]

(f) Can the following set of points be shattered by the set of all circles in $\mathbb{R}^2$? Justify your answer. [4]



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